Part I

Lab Assignment-8 (Introduction to Object-Oriented Programming)

Department: CSE **Subject: OOAD Lab (22CPP209)**

Objective: Understanding *Inheritance*.

Question 1. Package-delivery services, such as FedEx®, DHL® and UPS®, offer a number of different shipping options, each with specific costs associated. Create an inheritance hierarchy to represent various types of packages. Use class Package as the base class of the hierarchy, then include classes **TwoDayPackage** and **OvernightPackage** that derive from Package. Base class Package should include data members representing the name, address, city, state and ZIP code for both the sender and the recipient of the package, in addition to data members that store the weight (in ounces) and cost per ounce to ship the package. Package's constructor should initialize these data members. Ensure that the weight and cost per ounce contain positive values. Package should provide a public member function **calculateCost** that returns a double indicating the cost associated with shipping the package. Package's **calculateCost** function should determine the cost by multiplying the weight by the cost per ounce. Derived class **TwoDayPackage** should inherit the functionality of base class Package, but also include a data member that represents a flat fee that the shipping company ~~charges for two-day delivery service.~~ **TwoDayPackage**'s constructor should receive a value to initialize this data member. **TwoDayPackage** should redefine member function calculate- Cost so that it computes the shipping cost by adding the flat fee to the weight-based cost calculated by base class Package's **calculateCost** function. Class **OvernightPackage** should inherit directly from class Package and contain an additional data member representing an additional fee per ounce charged for overnight-delivery service. **OvernightPackage** should redefine member function **calculateCost** so that it adds the additional fee per ounce to the standard cost per ounce before calculating the shipping cost. Write a test program that creates objects of each type of Package and tests member function **calculateCost**.

Question 2 (Practice Question): Implement a graph data structure using suitable smart pointers by creating a class Node, which will create and initialize nodes with node labels, maintaining adjacent nodes and displaying the adjacent nodes to a node. Create another class graph that provides methods for adding nodes and the edges between the nodes, as well as showing the whole graph. Demonstrate the problem of cyclic reference and how your proposed solution handles it. [Hint: use the concept of a friend class to grant access to private data members.]